

Unit 8 Progress Check: FRQ

Name _____

1. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about weak acids. Use the K_a values in the following table to help answer some of the questions.

Acid	K_a
HF	6.7×10^{-4}
$\text{HC}_2\text{H}_3\text{O}_2$	1.8×10^{-5}

- (a) Using the expression $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$, explain how to determine which solution has the lower pH, 0.10 M HF(aq) or $0.10\text{ M HC}_2\text{H}_3\text{O}_2\text{(aq)}$. Do not perform any numerical calculations.



Please respond on separate paper, following directions from your teacher.

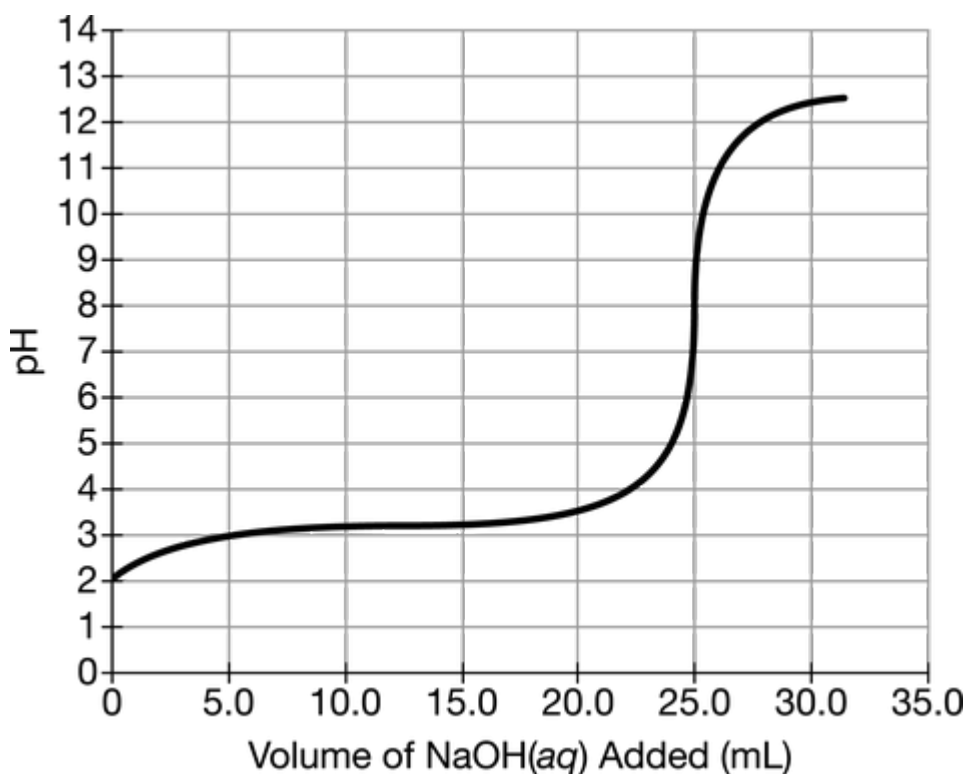
- (b) Which solution has a higher percent ionization of the acid, a 0.10 M solution of $\text{HC}_2\text{H}_3\text{O}_2\text{(aq)}$ or a 0.010 M solution of $\text{HC}_2\text{H}_3\text{O}_2\text{(aq)}$? Justify your answer including the calculation of percent ionization for each solution.



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(c) A student titrated **50.0 mL** of a **0.10 M** solution of a certain weak acid with **NaOH(aq)**. The results are given in the graph above.

(i) What is the approximate $\text{p}K_a$ of the acid?



Please respond on separate paper, following directions from your teacher.

(ii) What specific value from the graph is needed, in addition to the information in part (c) above, to calculate the molar concentration of the **NaOH(aq)**?



Please respond on separate paper, following directions from your teacher.

(iii) Identify a pH value between 2.5 and 7.5 at which the concentration of the weak acid being titrated is less than the concentration of its conjugate base.



Please respond on separate paper, following directions from your teacher.



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(d) In another experiment, the student titrated **50.0 mL** of **0.100 M HC₂H₃O₂** with **0.100 M NaOH(aq)**. Calculate the **pH** of the solution at the equivalence point.



Please respond on separate paper, following directions from your teacher.